

QUALITY OF AN APPROXIMATION

Let U be the unitary we want to implement. Then we say that V approximates U up to an error $\epsilon > 0$ if

$$E(U, V) \equiv \max_{|4\rangle} \|(U - V)|4\rangle\| < \epsilon$$

Prop The set $\{H, T\}$ is universal for 1 qubit gates up to an error ϵ .

Proof. HTH can be used to realize any rotation about $R_{\vec{n}}$. Then use the fact that for any $\epsilon > 0$ and $\vartheta \in [0, 2\pi)$, $E(R_{\vec{n}}^m(\beta), R_{\vec{n}}(\vartheta)) < \epsilon$

More precisely, we can show that

Exercise 4.3.2 Let $R_m(\theta_1), R_m(\theta_2)$ be 1-qubit rotations about the same axis.

(a) Show that distance between $R_m(\theta_1)$ and $R_m(\theta_2)$ satisfies $E(R_m(\theta_1), R_m(\theta_2)) \leq |e^{i\theta_1} - e^{i\theta_2}| \leq |\theta_1 - \theta_2|$.

(b) Let $\beta \in [0, 2\pi)$ is such that $\frac{\beta}{\pi}$ is not rational. Prove that for any $\varepsilon > 0$, and for any $\theta \in [0, 2\pi)$, there exists an integer n such that $E(R_m^n(\beta), R_m(\theta)) \leq \varepsilon$.

Hint: Use the pigeon-hole principle, which states the following. If $N > M$, then partitioning N elements into M disjoint sets, gives at least one set with more than 1 element.

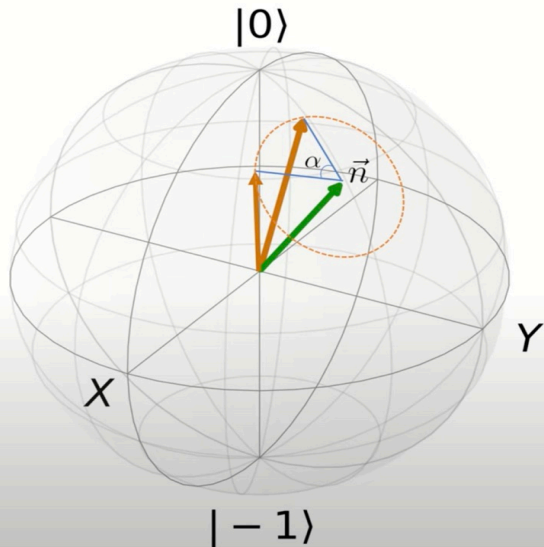
$$U = e^{i\delta} R_{\vec{n}}(\alpha)$$

$$R_{\vec{n}}(\alpha) = e^{-i \frac{\alpha}{2} \vec{n} \cdot \vec{\sigma}}$$

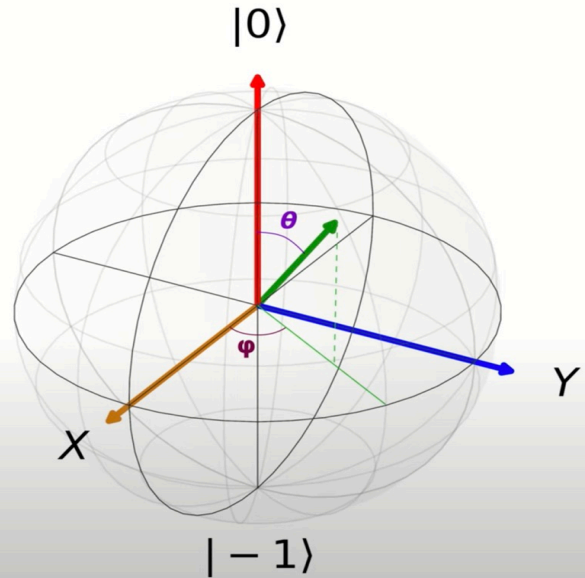
$$\alpha, \delta \in \mathbb{R}, \quad \vec{n} = (n_1 \sigma_x, n_2 \sigma_y, n_3 \sigma_z)$$

$$\vec{n} = (n_x, n_y, n_z)$$

$$\hat{R}_y(\theta) = e^{-i \frac{\theta}{2} \hat{Y}}$$



$$\hat{R}_z(\theta) = e^{-i \frac{\theta}{2} \hat{Z}}$$



Example : Realization of H

Arbitrary single qubit unitary operator

$$\hat{U} = e^{i\gamma} \hat{R}_{\vec{n}}(\alpha)$$

$$\hat{R}_n(\alpha) = e^{-i\frac{\alpha}{2}\vec{n}\cdot\vec{\sigma}} = \cos\frac{\alpha}{2} \hat{I} + i \sin\frac{\alpha}{2} \vec{n} \cdot \vec{\sigma}$$

$$\gamma = \frac{\pi}{2}, \alpha = \pi, \vec{n} = \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right)$$

$$\hat{U} = e^{i\frac{\pi}{2}} \left[\cos\frac{\pi}{2} \hat{I} - i \sin\frac{\pi}{2} \frac{1}{\sqrt{2}} (\hat{X} + \hat{Z}) \right] = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \hat{H}$$

(U2) Th. $\{H, T, CNOT\}$ approximates any circuits U :
$$\|\tilde{U} - U\| \leq \epsilon \quad \text{for some } \epsilon \geq 0$$

(where for a linear operator L , $\|L\| = \sup_{\|x\|=1} \|Lx\|$)

We first show that $\{H, T\}$ is universal for the operations on 1 qubit -

Preliminary result:

$HTH = R_x(\pi/4)$ up to a global phase factor

$$T = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix} = e^{i\pi/8} \begin{bmatrix} e^{-i\pi/8} & 0 \\ 0 & e^{i\pi/8} \end{bmatrix} = e^{i\pi/8} R_z(\pi/4)$$

$$HTH = e^{i\pi/8} H e^{-i\pi/4} Z H = e^{i\pi/8} e^{-i\pi/4} H Z H = e^{i\pi/8} R_x(\pi/4)$$

Using this result, we have:

$$\begin{aligned} HTHT &= e^{-i\frac{\pi}{8}\sigma_x} e^{-i\frac{\pi}{8}\sigma_z} = \\ &= \left(\cos \frac{\pi}{8} I - i \sin \frac{\pi}{8} \sigma_x \right) \left(\cos \frac{\pi}{8} I - i \sin \frac{\pi}{8} \sigma_z \right) = \\ &= \cos^2 \frac{\pi}{8} I - i \cos \frac{\pi}{8} \sin \frac{\pi}{8} \sigma_z - i \sin \frac{\pi}{8} \cos \frac{\pi}{8} \sigma_x - \sin^2 \frac{\pi}{8} \sigma_z \\ &= \cos^2 \frac{\pi}{8} I - i \left(\cos \frac{\pi}{8} (\sigma_z + \sigma_x) \right) \sin \frac{\pi}{8} - i^2 \sin^2 \frac{\pi}{8} \sigma_z \\ &= \cos^2 \frac{\pi}{8} I - i \left(\cos \frac{\pi}{8} (\sigma_z + \sigma_x) + \sin \frac{\pi}{8} \sigma_x \sigma_z \right) \sin \frac{\pi}{8} \end{aligned}$$

This is a rotation about $\vec{n} = \left(\cos \frac{\pi}{8}, \sin \frac{\pi}{8}, \cos \frac{\pi}{8} \right)$

Solovay-Kitaev Theorem

The set $\{CNOT, H, T\}$ is universal for quantum computation, that is, arbitrary quantum gates can be approximated up to an exponentially small error ε with only a polynomial overhead.

More precisely, the complexity of the simulation is $\mathcal{O}(\log \frac{K}{\varepsilon})$ for $K \in \mathbb{N}$.

The proof of this theorem is not part of the course, and can be found in Appendix 3 of the Chuang-Nielson textbook.

Notice that such decompositions with as efficient as possible overhead terms is an ongoing and important research topic.